

Approach:

- 1.) Insert a clone node after its every original node.
- 2.) In this way clone nodes and original one will interleave.
- 3.) Now set the random pointer of every clone (original) to random pointer of corresponding clone-original node.
- 4.) Separate the original and clone lists.
- 5.) Then original list will remain unchanged.

Pseudocode:

If (Head == NULL); return null;

For every original node:

create its clone.

insert clone immediately after original.

For every original node.

If original.random is not Null:

clone.random = original.random->next

Separate original list and clone list.

Return head of cloned.

Date: _____

Day: _____

Time Complexity: $O(N)$.

Space Complexity: $O(N)$.

Requirement Tests:

Test 1

Empty List

Expected clone
= Null.

Test 2

$1 \rightarrow 2 \rightarrow 3$.

All Random = NULL

Expected clone

$1 \rightarrow 2 \rightarrow 3$.

All clone random
= NULL

Test 3

$1 \rightarrow 2 \rightarrow 3$.

Node 1 Random $\rightarrow$ Node 1

Node 2 Random $\rightarrow$ Node 2

Node 3 Random $\rightarrow$ Node 3

Expected

Each clone node's
random points

to its own clone.